HESTON'S STOCHASTIC VOLATILITY MODEL

Stochastic volatility models extend constant and local volatility models by allowing the variance itself to evolve randomly over time. This added source of uncertainty enables stochastic volatility models to capture key empirical features such as volatility clustering, persistent regimes of high or low variance, and the skew observed in implied volatility surfaces. Among these models, the Heston model is one of the most widely used and influential.

5.1 Asset Dynamics

Heston's model assumes that the underlying asset, S, follows a diffusion with a stochastic variance, v, whilst the variance itself is modelled as a mean-reverting square-root process. To continue with the same mathematical terms, r is the risk-free interest rate, κ is the speed of mean reversion, θ is the long-run variance, ξ is the volatility of variance, ρ is the correlation between asset returns and variance shocks, and dX_1 and dX_2 are random variables following a Brownian motion.

First, let us understand the asset dynamics of the Heston stochastic volatility model under the real world probability measure. The dynamics are given by:

$$\frac{dS}{S} = \mu dt + \sqrt{v} S dX_1^{\mathbb{P}}, \tag{18}$$

$$dv = \kappa\left(\theta - v\right) dt + \xi\sqrt{v} dX_2^{\mathbb{P}}, \tag{19}$$

$$\rho dt = dX_1^{\mathbb{P}} dX_2^{\mathbb{P}}. \tag{20}$$

Using Girsanov's Theorem [Girsanov 1960] to transform $\mathbb{P}$ to $\mathbb{Q}$ (*i.e.*, transforming from the real world probability measure to the risk-neutral probability measure), see Appendix B:

$$dX_S^{\mathbb{Q}} = dX_S^{\mathbb{P}} + \alpha_S dt,$$
$$dX_v^{\mathbb{Q}} = dX_v^{\mathbb{P}} + \alpha_v dt,$$

where:

$$\alpha_S = \frac{\mu^{\mathbb{P}} - r}{\sqrt{v}}, \qquad \alpha_v = \frac{\lambda}{\xi^{\mathbb{P}}}\sqrt{v},$$

where λ is the variance risk premium. Thus, the asset dynamics under the risk-neutral measure are:

$$\frac{dS}{S} = r dt + \sqrt{v} S dX_1^{\mathbb{Q}}, \tag{21}$$

$$dv = \kappa^{\mathbb{Q}}\left(\theta^{\mathbb{Q}} - v\right) dt + \xi\sqrt{v} dX_2^{\mathbb{Q}}, \tag{22}$$

$$\rho dt = dX_1^{\mathbb{Q}} dX_2^{\mathbb{Q}}, \tag{23}$$

where $\rho^{\mathbb{Q}} = \rho$, $\kappa^{\mathbb{Q}} = \kappa + \lambda$, and $\theta^{\mathbb{Q}} = \frac{\kappa\theta}{(\kappa+\lambda)}$.

5.2 Derivation of the Fair Value of a European Call Option

Using standard arbitrage arguments, under the Heston model, we arrive at Garman's partial differential equation, which is expressed in terms of the price, S:

$$\frac{\partial C}{\partial t} + \frac{1}{2} v S^2 \frac{\partial^2 C}{\partial S^2} + \rho\xi v S \frac{\partial^2 C}{\partial v \partial S} + \frac{1}{2}\xi^2 v \frac{\partial^2 C}{\partial v^2} + rS\frac{\partial C}{\partial S} + \left[\kappa(\theta - v) - \lambda v\right]\frac{\partial C}{\partial v} - rC = 0. \tag{24}$$

Heston builds the solution of the PDE above by the method of characteristic functions. He looks for a solution of the form corresponding to the Black-Scholes model.

$$C(S_0, E, v_0, \tau) = S_0 P_1 - E e^{-r\tau} P_2. \tag{25}$$

Where P_1 is the Δ of the European call option, and P_2 is the conditional risk neutral probability that the asset price will be greater than the strike price, E, at maturity. Both probabilities, P_1, P_2, also satisfy the PDE provided that characteristics functions, φ_1, φ_2, are known in terms of P_1, P_2 and are defined via the inverse Fourier transform. If you manipulate the PDE by $X = \ln(S)$. We can arrive at:

$$P_j = \frac{1}{2} + \frac{1}{\pi}\int_0^\infty \Re\left[\frac{e^{-i\phi \ln E}\varphi_j\left(X_0, E, v_0, \tau; \phi\right)}{i\phi}\right] d\phi, \qquad j \in \{1, 2\}. \tag{26}$$

Heston assumes the characteristic functions, φ_1, φ_2, have the form:

$$\varphi_j\left(X_0, E, v_0, \tau; \phi\right) = e^{C(\tau;\phi)+D(\tau;\phi)v+i\phi X}. \tag{27}$$

If you solve the PDE, then we are left with the semi-analytical solutions:

$$C\left(\tau;\phi\right) = r\phi i\tau + \frac{a}{\xi^2}\left[\left(b_j - \rho\phi i - d\right)\tau - 2\ln\left(\frac{1-ge^{-d\tau}}{1-g}\right)\right], \tag{28}$$

$$D\left(\tau;\phi\right) = \frac{b_j - \rho\xi\phi i - d}{\xi^2}\left(\frac{1-e^{-d\tau}}{1-ge^{-d\tau}}\right), \tag{29}$$

where:

$$g = \frac{b_j - \rho\xi\phi i - d}{b_j - \rho\xi\phi i + d}, \qquad d = \sqrt{\left(\rho\xi\phi i - b_j\right)^2 - \xi^2\left(2u_j\phi i - \phi^2\right)},$$
$$u_1 = \frac{1}{2}, \qquad u_2 = -\frac{1}{2}, \qquad a = \kappa\theta, \qquad b_1 = \kappa + \lambda - \rho\xi, \qquad b_2 = \kappa + \lambda.$$

Let us obtain a simplified equation for C with one integral, here, we add the Fourier integral into the European call option equation. Note, from (30) to (31), we can take S and the discounted strike factor out.

$$C\left(S_0, E, v_0, \tau\right) = S_0\left[\frac{1}{2} + \frac{1}{\pi}\int_0^\infty \Re\left[\frac{e^{-i\phi\ln E}\varphi_1\left(X_0, E, v_0, \tau; \phi\right)}{i\phi}\right]d\phi\right]$$
$$- Ee^{-r\tau}\left[\frac{1}{2} + \frac{1}{\pi}\int_0^\infty \Re\left[\frac{e^{-i\phi\ln E}\varphi_2\left(X_0, E, v_0, \tau; \phi\right)}{i\phi}\right]d\phi\right] \tag{30}$$

$$= \frac{1}{2}\left(S_0 - Ee^{-r\tau}\right) + \frac{1}{\pi}\int_0^\infty \Re\left[S\frac{\varphi_1\left(\phi\right)}{i\phi E^{i\phi}} - Ee^{-r\tau}\frac{\varphi_2\left(\phi\right)}{i\phi E^{i\phi}}\right]d\phi \tag{31}$$

$$= \frac{1}{2}\left(S_0 - Ee^{-r\tau}\right) + \frac{1}{\pi}\int_0^\infty \Re\left[e^{r\tau}\left(\frac{\varphi\left(\phi - i\right)}{i\phi E^{i\phi}} - E\frac{\varphi\left(\phi\right)}{i\phi E^{i\phi}}\right)\right]d\phi. \tag{32}$$

From (31) to (32), we have now one integral in just φ, and changed the parameters, d and g, so that they are not dependent on b_1, b_2, u_1 or u_2. Where:

$$\varphi\left(X_0, E, v_0, \tau; \phi\right) = e^{r\phi i\tau}S_0^{i\phi}\left(\frac{1-ge^{-d\tau}}{1-g}\right)^{-\frac{2a}{\xi^2}}$$
$$\exp\left[\frac{a\tau}{\xi^2}\left(b_2 - \rho\xi\phi i - d\right) + \frac{v_0}{\xi^2}\left(b_2 - \rho\xi\phi i - d\right)\left(\frac{1-e^{-d\tau}}{1-ge^{-d\tau}}\right)\right], \tag{33}$$

where:

$$g = \frac{b - \rho\xi\phi i - d}{b - \rho\xi\phi i + d}, \qquad d = \sqrt{\left(\rho\xi\phi i - b\right)^2 + \xi^2\left(\phi i + \phi^2\right)}, \qquad a = \kappa\theta, \qquad b = \kappa + \lambda.$$

We can rearrange the solution, and work out why these parameters are no longer featured. Recall (27), and remember that we can utilise that $X = \ln\left(S_0\right)$, therefore, $e^{i\phi X} = e^{\ln\left(S^{i\phi}\right)} = S^{i\phi}$. So, we can take that down outside of the integral. Taking the exponential of $C\left(\tau;\phi\right)$, (28):

$$e^{C(\tau;\phi)} = e^{r\phi i\tau}\exp\left[\frac{a}{\xi^2}\left(b_j - \rho\xi\phi i - d\right)\tau\right]\exp\left[\ln\left(\frac{1-ge^{-d\tau}}{1-g}\right)^{-\frac{2a}{\xi^2}}\right] \tag{34}$$

$$= e^{r\phi i\tau}\left(\frac{1-ge^{-d\tau}}{1-g}\right)^{-\frac{2a}{\xi^2}}\exp\left[\frac{a\tau}{\xi^2}\left(b_j - \rho\xi\phi i - d\right)\right] \tag{35}$$

For both d_j and g_j, to reconcile the difference between b_1 and b_2 calculations, we can see that this is taken into account by the offset of $\varphi(\phi - i)$ compared to $\varphi(\phi)$. To understand the change in the b_j term to b, look at the first term of d, if we substitute $\Phi = \phi - i$ into the function $\varphi(\Phi)$. Thus:

$$\rho\xi\left(\phi - i\right)i - b = \rho\xi\left(\phi i - i^2\right) - \left(\kappa + \lambda\right) = \rho\xi\left(\phi i + 1\right) - \left(\kappa + \lambda\right)$$
$$= \rho\xi\phi i - \left(\kappa + \lambda - \rho\xi\right) = \rho\xi\phi i - b_1,$$
$$\rho\xi\phi i - b = \rho\xi\phi i - \left(\kappa + \lambda\right) = \rho\xi\phi i - b_2.$$

To understand the change in the u_j term, now look at the second term of d and complete the same substitution:

$$\xi^2\left(\phi i+\phi^2\right)=\xi^2\left((\phi-i)\,i+(\phi-i)^2\right)=\xi^2\left(\phi i+1+\phi^2-2\phi i-1\right)$$
$$=\xi^2\left(-\phi i+\phi^2\right)=-\xi^2\left(2u_1\phi i-\phi^2\right)=-\xi^2\left(2\left(\frac{1}{2}\right)\phi i-\phi^2\right),$$
$$\xi^2\left(\phi i+\phi^2\right)=-\xi^2\left(2u_2\phi i-\phi^2\right)=-\xi^2\left(2\left(-\frac{1}{2}\right)\phi i-\phi^2\right).$$

5.3 Heston Parameters to Determine via Calibration with Market Prices

Now that we have the function to determine the Heston prices for each option chain, we need to determine the unknown parameters of the Heston model. Let $\Theta=(v_0,\kappa,\theta,\xi,\rho,\lambda)$. The aim of the calibration is to minimise the ordinary least squared errors [Mikhailov and Nögel 2003]:

$$SqErr(\Theta)=\sum_{i=1}^{N}\sum_{j=1}^{M}w_{ij}\left[C_{MP}\left(X_i,\tau_j\right)-C_{SV}\left(S_\tau,X_i,\tau_j,r_j,\Theta\right)\right]^2+\text{Penalty}\left(\Theta,\Theta_0\right). \tag{36}$$

The penalty function may be, *e.g.*, the distance to the initial parameter vector, $\text{Penalty}\left(\Theta,\Theta_0\right)=\parallel\Theta-\Theta_0\parallel^2$, but for simplicity, in these calculations, we make the penalty function to be zero. All strikes and tenures on the FTSE 100 options index are quite liquid and we will weigh them the same (*i.e.*, $\frac{m}{n}$). So, for every market price, we will calculate the Heston price with a set of parameters. The optimisation objective function thus becomes:

$$\hat{\Theta}=\underset{\Theta\in U_\Theta}{\arg\min}\,SqErr\left(\Theta\right). \tag{37}$$

Here, we assume that the set of possible combinations of parameters, U_Θ, is compact and in the range for which a solution exists.

5.4 Plotting the Heston Implied Volatility Surface

Now that we have calculated all Heston prices for all strikes and maturities for European call options. Then, using the Black-Scholes model, we can minimise the difference between the prices using the Newton-Raphson method to find the Heston implied volatility and model the surface. Figure (1) displays the Heston output of the Python code. See Appendix A for all code on calculating the fair value and plotting the implied volatility surface.

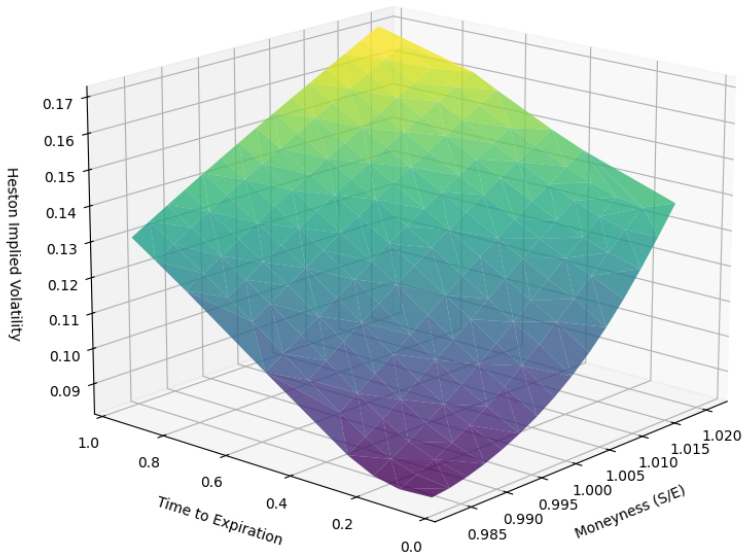


Figure 1: Heston Implied Volatility Surface.

5.5 Mathematical Properties of the Heston Model

The Heston model introduces a stochastic variance process with mean reversion, ensuring volatility fluctuates realistically and stabilises around a long-run level. The square root diffusion helps maintain positivity in implied volatility. The correlation parameter, ρ, plays a critical role in shaping the implied volatility skew, where

negative correlation between asset returns and variance shocks generates higher implied volatilities for lower strikes, which is consistent with equity market data. Its closed-form characteristic function enables efficient option pricing and calibration via Fourier methods.

In practice, allowing volatility to evolve stochastically makes the Heston model more suitable for risk management than constant or local volatility models, as it captures volatility clustering, variance risk premia, and provides more realistic hedging exposures for both vanilla and exotic derivatives. However, the model introduces its own sources of model risk (the risk arising from incorrect model assumptions, unstable parameters, or poor calibration), particularly through parameter instability (the tendency for a model's calibrated parameters to fluctuate unpredictably as market conditions change) and calibration sensitivity (the degree to which a model's behaviour or prices are affected when its calibrated parameters are adjusted).